Classifier Based on the LightGBM Model

Team Name: **Super Learning**

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1 Introduction

To predict users' social roles based on their tweet content, we first analyzed the structure of the dataset, designed our own feature engineering pipeline, and implemented a prediction model using LightGBM. Our solution achieved a competitive accuracy score on Kaggle.

2 Data Processing – Feature Selection

To facilitate data analysis, we first examined all features in the `train.jsonl` file and performed an initial, commonsense-based assessment to identify which features were likely relevant to our task.

Since our focus was on predicting the identity label of the user who posted the tweet, we excluded all fields under `quoted_status.*`. Additionally, we discarded low-value features such as tweet ID, geographic location, and language. We identified the following categories of features as particularly important:

User-related fields (`user.*`)

Category	Field	Description
Core identity signals	`user.description`	Bio text → useful for extracting profession-related keywords (e.g., *médecin*, *journaliste*)
	`user.url`	Professional users are more likely to include an official website
Influence indicators	`user.listed_count`	Number of times listed in public lists → indicator of high-quality accounts
Behavior & activity	`user.statuses_count`	Total number of tweets posted
	`user.created_at`	Account creation time → used to compute `account_age_days`
Account quality	`user.default_profile`	Uses default profile/background → common among bots
	`user.default_profile_image`	Same as above
	`user.protected`	Private account? → Influencers are typically public

Text fields

Includes `text` and `extended_tweet.full_text`, which provide the raw tweet content for inferring user personas.

Engagement metrics

`retweet_count`, `quote_count`, `reply_count`, `favorite_count`: used to assess tweet influence and indirectly infer user credibility.

Entity fields (`entities.*`)

Field	Description
`entities.urls`	Whether the tweet contains links
`entities.media`	Whether the user frequently posts media (images/videos)

We focused primarily on these four categories of features. Discarding redundant or low-value fields helped reduce noise and mitigate overfitting risks.

3 Exploratory Data Analysis (EDA)

To build an efficient and interpretable model for detecting credible influencers, we conducted systematic exploratory data analysis (EDA) on the training set (`train.jsonl`).

3.1 Label Distribution

The dataset contains 53.37% non-influencers (label 0) and 46.63% influencers (label 1). This near-balanced distribution indicates that class resampling is unnecessary.

3.2 User Metadata Analysis

We first examined numerical user features after applying a log transformation to handle long-tailed distributions. The median of `log_user.listed_count` was approximately 3.5 for non-influencers and 4.5 for credible influencers, indicating that influencers are listed in public lists about three times more often. Similarly, the median of `log_user.statuses_count` was around 9.0 for non-influencers versus 10.5 for influencers, reflecting higher activity levels. Account age (in days, log-transformed) showed a modest difference: medians of roughly 8.1 versus 8.4, suggesting influencers tend to have slightly older accounts.

For categorical features, the presence of a personal or professional website (`user.url`) proved highly discriminative: among users with a URL, 75.17% were credible influencers, compared to only 31.26% among those without one. Conversely, users with a default profile (`user.default_profile = True`) were much less likely to be influencers—only 33.30% of such users belonged to class 1, versus 66.26% for those with customized profiles. However, the fields `user.default_profile_image` and `user.protected` were uniformly `False` across all samples, providing no predictive signal, and were therefore excluded from modeling.

3.3 Bio Keyword Analysis

We analyzed the frequency of profession-related keywords in user bios. The keyword indicating a doctor (`is_doctor`) appeared in 1.89% of non-influencer bios but in 4.06% of influencer bios—a 2.14-fold increase. The journalist indicator (`is_journalist`) showed an even stronger association: it occurred in 2.72% of non-influencer bios versus 7.70% of influencer bios, representing a 2.83× lift. Most notably, the "official" identity marker (`is_official`) was present in 1.84% of non-influencer bios but in 6.96% of influencer bios, yielding a 3.78× enrichment. These results confirm that bio content provides valuable signals for distinguishing credible influencers.

3.4 Tweet Behavior Analysis

At the tweet level, we evaluated structural features related to entities. Tweets containing media (`has_media = True`) were strongly associated with credible influencers: 64.03% of such tweets came from class-1 users, compared to 46.03% for tweets without media—an 18 percentage point difference. The presence of hashtags also showed predictive power: 59.36% of hashtag-containing tweets originated from influencers, versus 44.10% for those without—a gap of 15.26 percentage points. In contrast, including a URL had negligible impact, with influencer probabilities of 47.02% (with URL) versus 46.02% (without).

We also inspected engagement metrics—retweet count, quote count, reply count, and favorite count—but found that their log-transformed medians were zero for both classes. This suggests the majority of tweets received no interactions, rendering these features ineffective for modeling purposes.

3.5 Final Selected Features

Based on the above findings, we selected the following features for our modeling pipeline: `user.url`, `user.listed_count`, `user.statuses_count`, `has_media`, `has_hashtag`, along with keyword-derived indicators `is_journalist`, `is_doctor`, `is_official`, and `user.default_profile`.

4 Feature Engineering

4.1 Data Preprocessing

- **Flattening**: Each line in `train.jsonl` and `kaggle_test.jsonl` is a JSON object representing one tweet. We used `json_normalize` to flatten nested structures (e.g., `user.*`, `extended_tweet.*`) into DataFrame columns.
- **Text Extraction**: We prioritized `extended_tweet.full_text` over `text` to ensure semantic completeness.

4.2 User Feature Extraction

Following EDA insights, we extracted:

- Boolean flags: whether `user.url`, `user.default_profile`, `entities.media`, and `entities.hashtags` are present (non-empty).
- Numerical features: `user.listed_count` and `user.statuses_count`, log-transformed using `log1p(·)` to handle long-tailed distributions.

- Identity signals: detected via keyword matching in `user.description`.

4.3 Additional Ratio Features

Feature	Construction	Interpretation
`account_age_days`	`now - user.created_at` (in days)	Older accounts are more trustworthy
`tweets_per_day`	`log1p(statuses / (age + 1))`	Daily tweeting frequency → activity intensity
`listed_per_status`	`(listed + 1) / (statuses + 1)`	"Influence density" — official recognition per tweet

4.4 Deep Semantic Features

Given that most tweets are in French, we used **CamemBERT**—a pre-trained language model specifically designed for French—to extract semantic embeddings:

- Generated 768-dimensional embeddings for both `full_text` and `user.description`.
- Concatenated these embeddings with other engineered features.
- Applied `StandardScaler` to numerical features to avoid disrupting the semantic structure of BERT embeddings.

5 Model Construction

5.1 Model Choice

We selected **LightGBM**, a high-performance, distributed gradient-boosting framework based on decision trees. It efficiently handles mixed feature types (sparse booleans, continuous numerics, high-dimensional embeddings), supports GPU acceleration, and is robust to noise.

5.2 Training & Optimization Strategies

Overfitting was the main challenge. To address it, we employed several techniques:

- **Early stopping**: monitored AUC; training halted if no improvement after 100 rounds.
- **Hyperparameter tuning**: reduced `num_leaves`, increased `min_child_samples`, adjusted `max_depth`, and used a lower learning rate for stable convergence.
- **Threshold tuning**: since positive samples constitute 46.63% (not 50%), we searched for the optimal classification threshold on the validation set.

5.3 Other Attempts (Discarded)

- **PCA on text embeddings**: did not improve performance—likely because 768 dimensions are already moderate, and further reduction lost semantic information.
- **Ensemble models**:
  - Soft voting between XGBoost and LightGBM.
  - Weighted ensemble of full-feature model and model without BERT features.

  Both ensembles converged toward the full LightGBM model and offered no gain, so they were abandoned.

6 Conclusion

Our final submission achieved a **Kaggle score of 0.845**. While this represents our current performance ceiling, there remains a gap compared to top submissions, indicating room for further learning and improvement—particularly in hyperparameter optimization and feature refinement.

Appendix

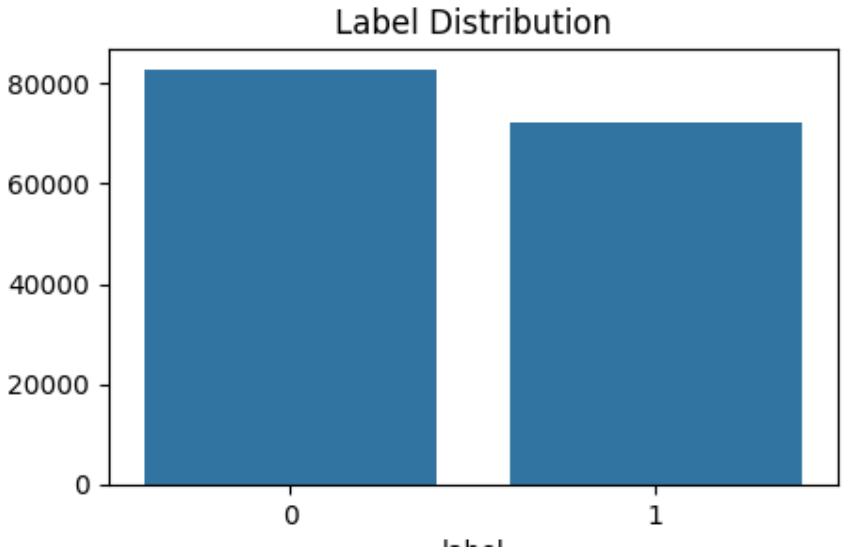

Figure 1: Distribution of labels: 0 (non-influencers) and 1 (influencers).

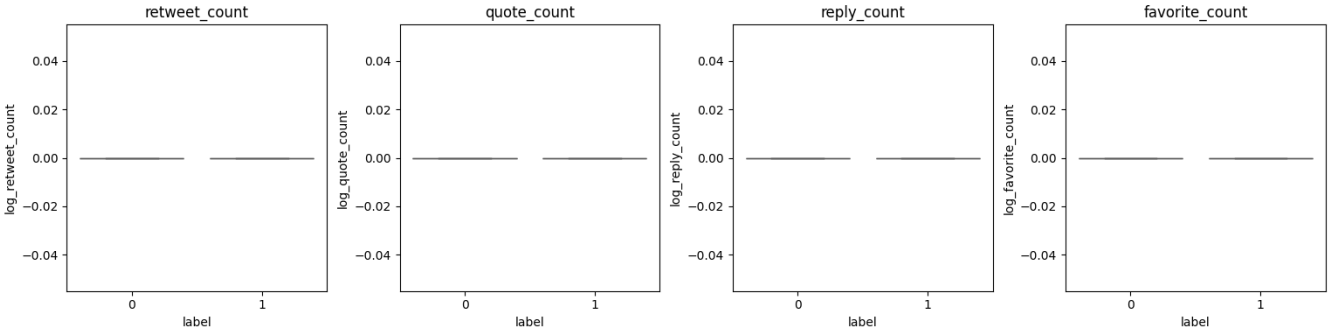

Figure 2: Log-transformed engagement counts by label: retweet, quote, reply, favorite.

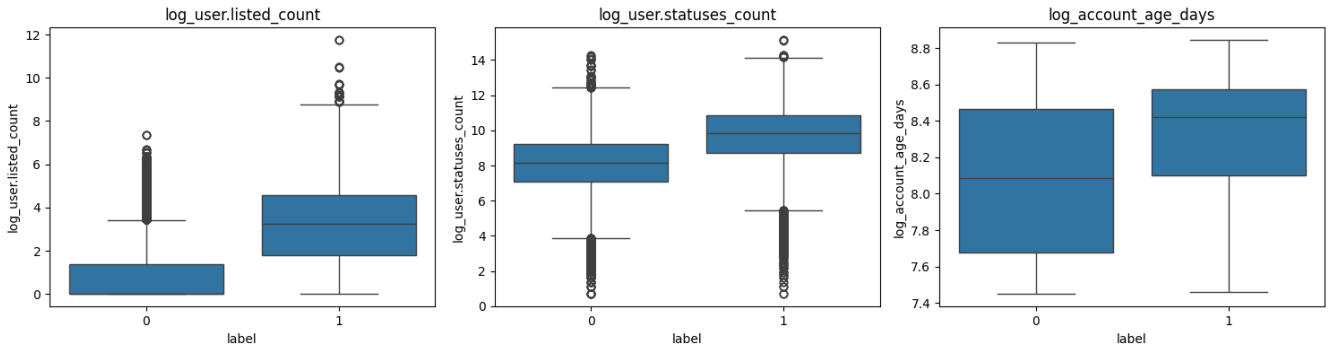

Figure 3: Boxplots of log-transformed user features: listed_count, statuses_count, account_age_days.

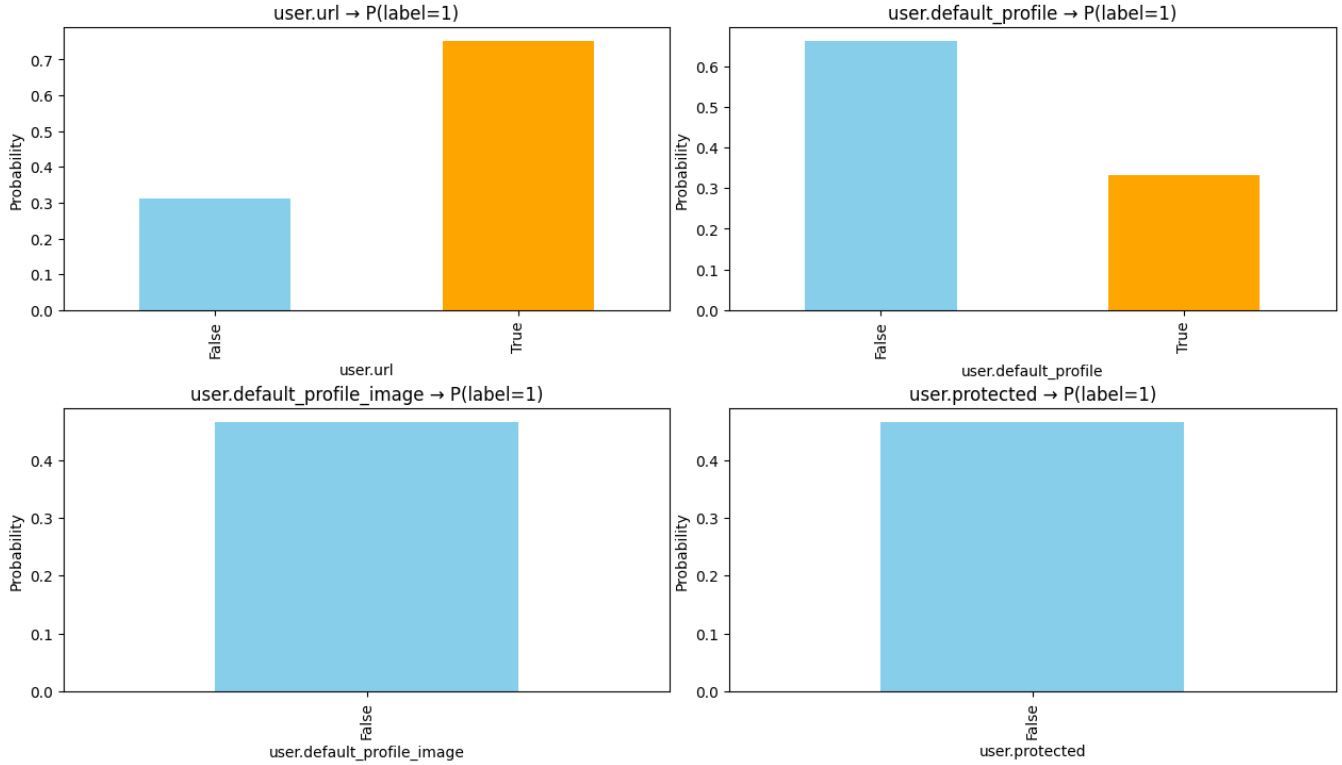

Figure 4: Probability of being an influencer given binary features: url, default_profile, default_profile_image, protected.

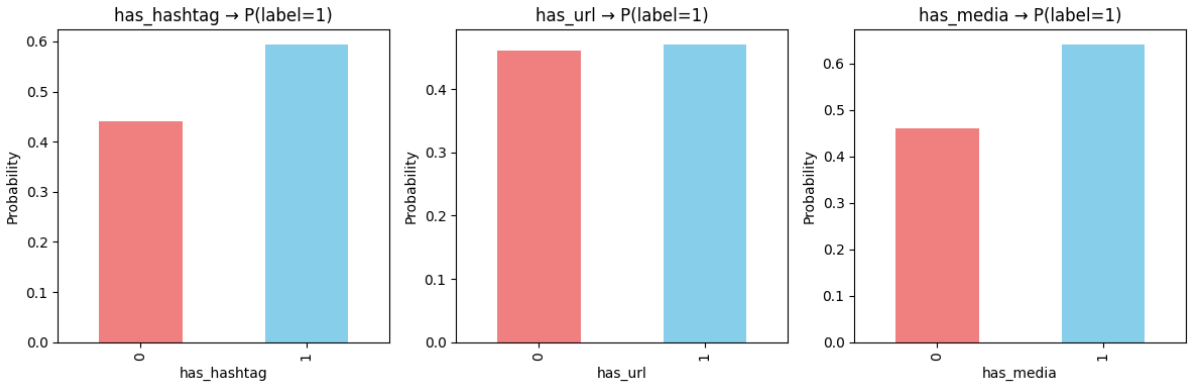

Figure 5: Hit rate of being an influencer given keywords: is_doctor, is_journalist, is_official.

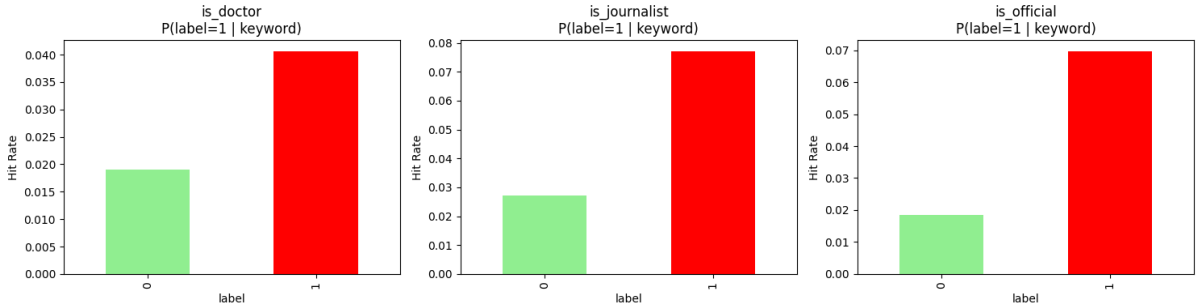

Figure 6: Probability of being an influencer given tweet entities: has_hashtag, has_url, has_media.